Team Details

- Team name: BYTEME
- Team leader name: Jyotier Vithalani
- Team leader email-id: Jyotier.v@ahduni.edu.in
- Problem Statement: Cyber-Resilient Infrastructure for Critical Sectors
- Team Members: Tanmay Kanani, Krishna Patel, Jinay Shah, Akshay Patel

~ BYTEME ~

Problem Overview

Critical digital infrastructures in healthcare, agriculture, and urban systems operate at large scale and are expected to be always available and secure. However, these **systems are increasingly exposed to sophisticated cyber threats**, many of which operate over encrypted traffic and evade traditional security mechanisms.

Our Idea

- An intelligent, behaviour-based security system that **detects zero-day and advanced threats hidden in encrypted traffic without decryption**.
- **Machine learning-driven** anomaly detection enables automated, **rapid threat containment** and delivers an adaptive, **future-ready cyber-resilience layer** for critical infrastructure.

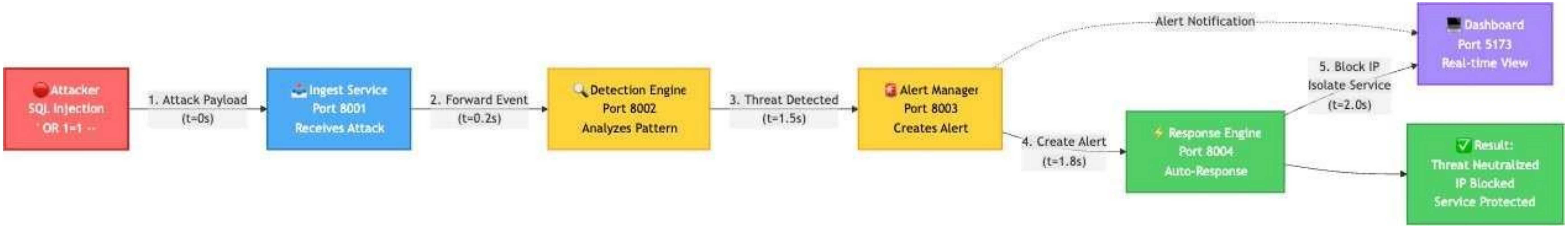
Our Progress

- Implemented kubernetees with horizontal pod scaling.
- Add email on alert/attack functionality.
- Added rate limiting (customisable value) in the node software.
- Added ip blocking in by node.
- Developed a unified setup utility that provisions the complete Kubernetes environment.
- Federated learning on telemetry data.

Proof of Concept

📋 Problem → Solution Mapping

Problem Requirement	Your Solution	Status
Real-time monitoring	Ingest Service + WebSocket	✅
Anomaly detection algorithms	ML Models (LSTM, RF) + Rule Engine	✅
Alerting system with severity levels	Alert Manager (low/medium/high/critical)	✅
Automated response mechanisms	Response Engine + Playbooks	✅
Dashboard showing security status	React Frontend	✅
Simulated attack scenarios	Attack Simulator Scripts	✅
Healthcare/Agriculture/Urban domains	Domain-specific ML models	✅
Scalability	Kubernetes + HPA	✅

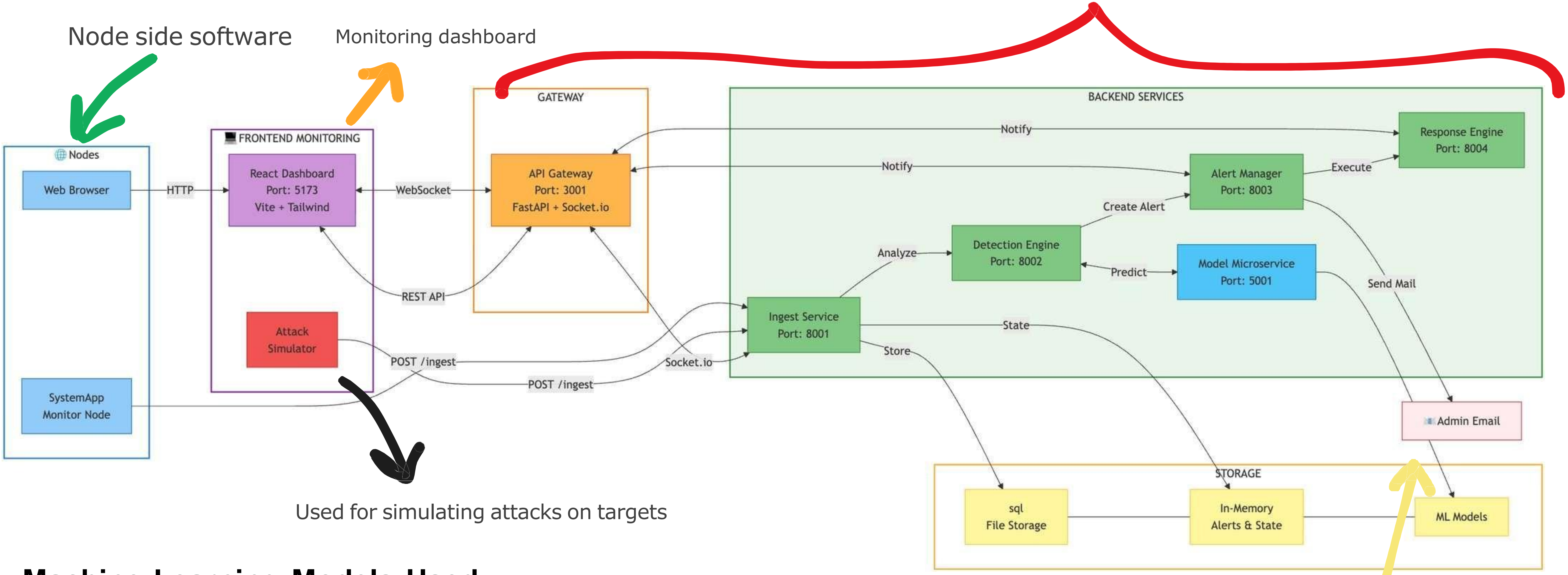


Approach

- Traffic Capture & Flow Construction
- **Network traffic is captured at the edge,** extracting timing, volume, and protocol-level metadata.
- Encryption-Aware Routing
- **Rule-Based Analysis for Unencrypted Traffic**
- Unencrypted traffic is inspected using predefined signatures and security rules for fast, deterministic detection.
- Behaviour-Based Analysis for Encrypted Traffic
- **Encrypted traffic bypasses payload inspection and is analysed using flow-level behavioural features.**
- Domain-Aware Model Selection
- The system dynamically selects the appropriate detection model based on the operational context:
  - General Network → **Neural Network Classifier**
  - Agriculture → **LSTM Autoencoder**
  - Healthcare → **LSTM Classifier**
  - Urban → **LSTM Forecaster**
- Threat Scoring & **Decision Engine**
- Automated & Manual Response Execution
- **Federated Learning Feedback Loop**
- **Local learnings are aggregated centrally and redistributed,**

Architecture Diagram

Each service has its own **docker** image which is then scaled on **kubernetees**

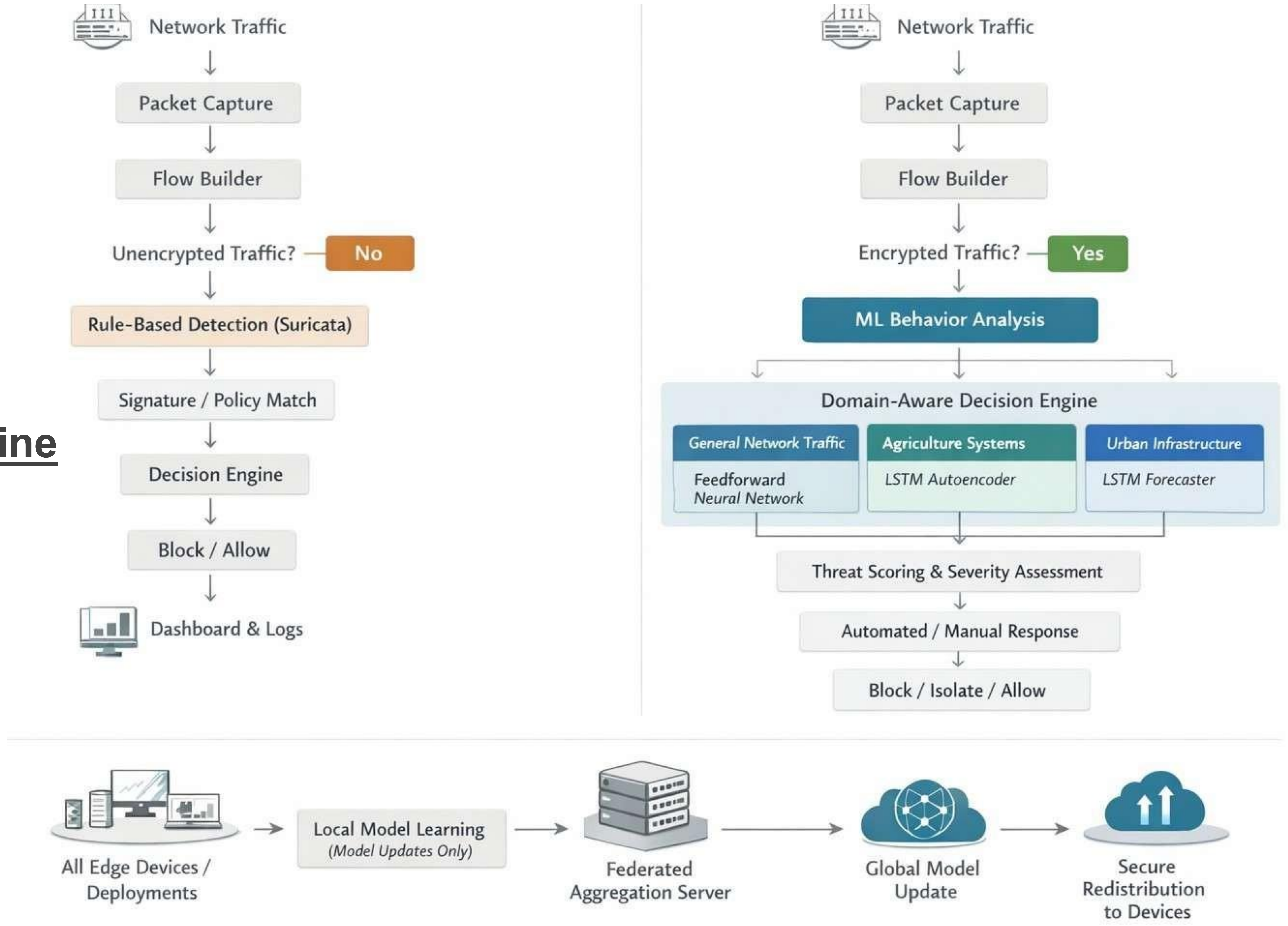


Email notification

Machine Learning Models Used

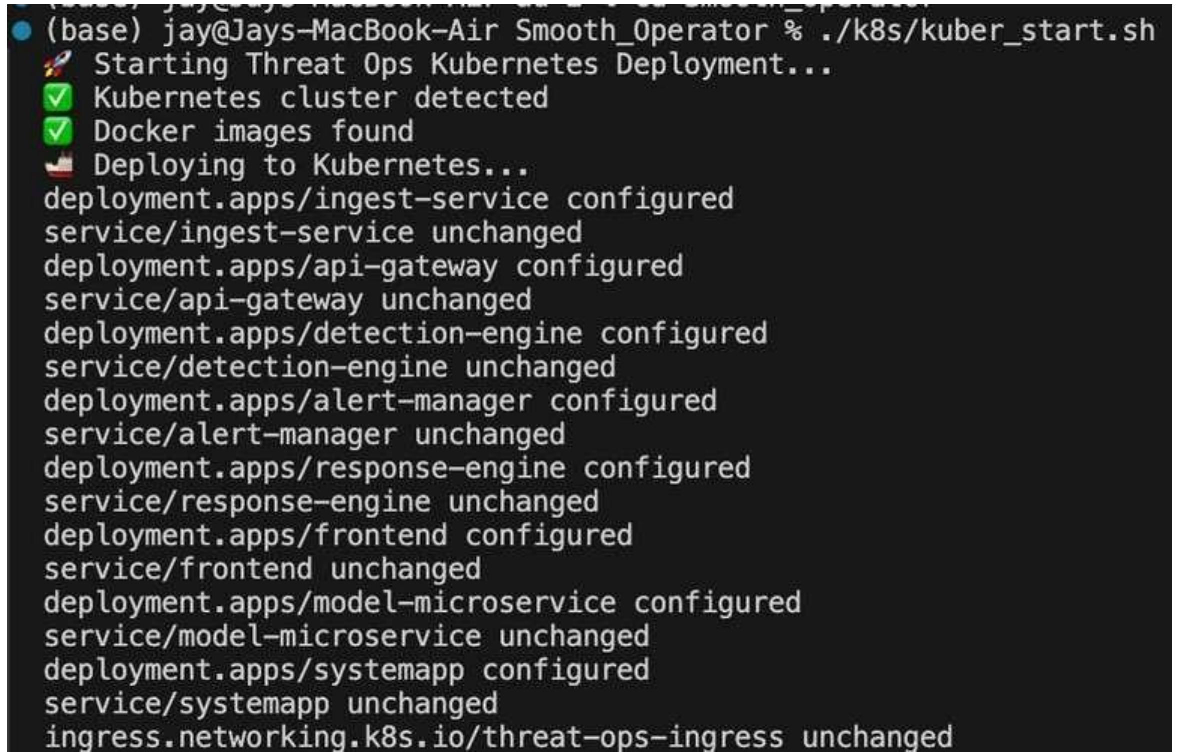
1. **Feedforward Neural Network** (PyTorch): **Generic network attack detection** (DDoS, Botnet, Brute Force)
2. **Isolation Forest** (Unsupervised ML): Network flow based anomaly detection on encrypted traffic
3. **LSTM Autoencoder** (Unsupervised): Agriculture sensor spoofing and physics-based anomaly detection
4. **LSTM Classifier** (Supervised): Healthcare IoMT DDoS and traffic surge detection
5. **LSTM Forecaster** (Regression) : Urban traffic flow prediction and grid manipulation detection

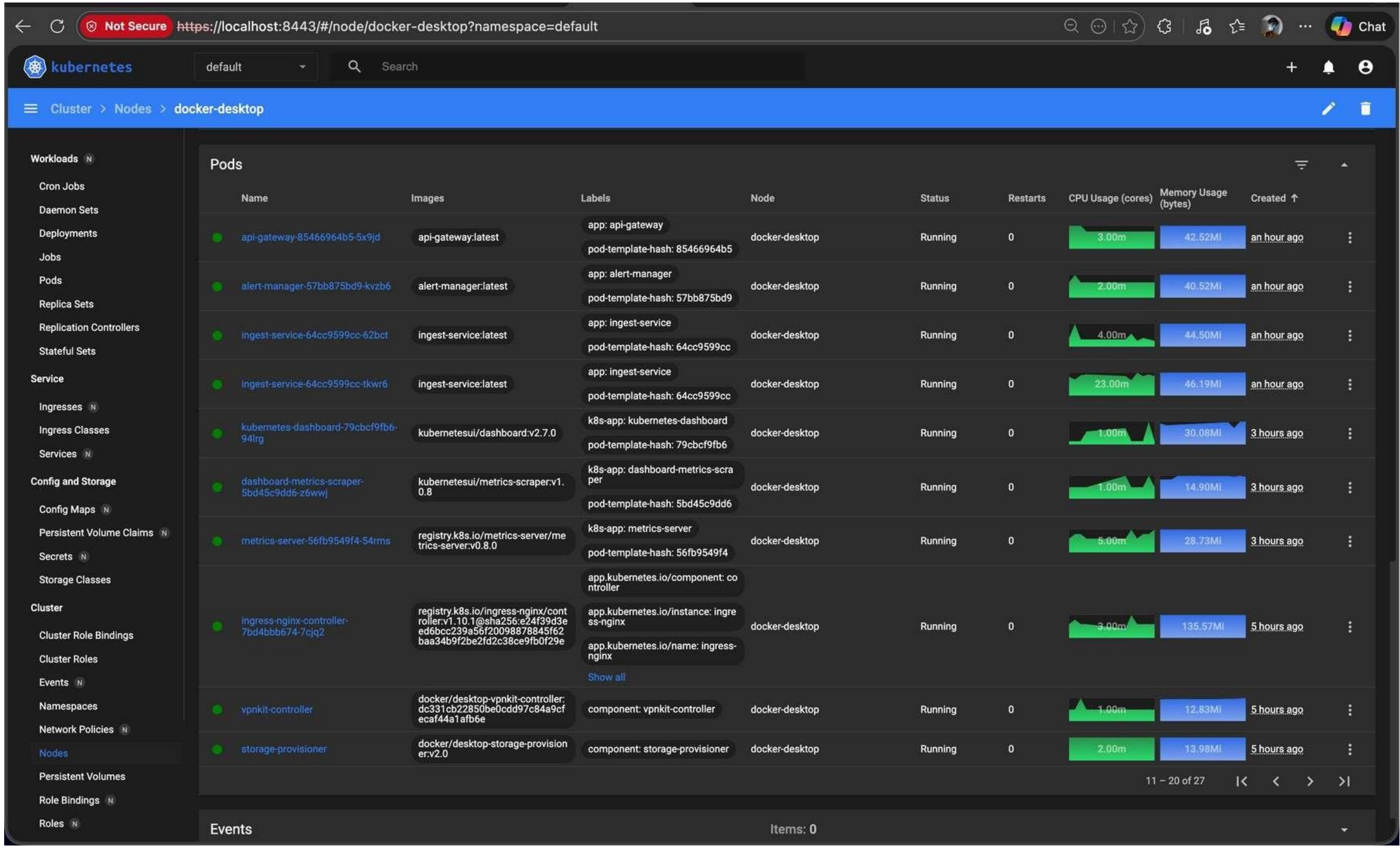
Detection engine



Scalability : Kubernetes

- The platform uses Kubernetes **Horizontal Pod Autoscaler** to dynamically scale services based on real-time resource utilization.
- **Self-Healing Infrastructure**: Failed or compromised pods are automatically recreated.
- **Multi-Replica Critical Services:** Detection, alerting, and dashboard services run with replicas.





PRojECt FEAtuRES

ZERo-DAY & EnCRYPtED AttACK DEtECtion

- **Live Packet Capture** : Uses **unsupervised machine learning** to learn effectively detects previously unseen attacks that bypass signature-based security systems

LAYERED AI DEfEnSE ARCHitECtuRE

- Ensures early-stage filtering of broad attacks while enabling **deep contextual analysis** for healthcare, agriculture, and urban systems.

FEDERAtED LEARninG-EnABlED CollECtivE DEfEnSE

- Only model updates and threat insights are shared and **aggregated centrally** to improve global detection accuracy.

KUBERNETES-BASED SCALABLE DEPLOYMENT

- Uses containerization and Kubernetes orchestration to automatically scale edge services and central components based on traffic load, ensuring high availability, fault tolerance, and consistent performance under attack conditions.

AUTOMATED RESPONSE & POLICY ENFORCEMENT (SOAR)

- Automatically blocks malicious IPs, and escalates alerts based on threat severity and confidence levels.

REAl-TimE DASHBoARD & ViSuAlizAtion

- Displays **live telemetry, alert timelines, device status**, and response actions through **WebSockets**.

AttACK SimulAtion & TEStinG FRAmEwoRK

- Includes **attacker and victim simulation** tools to validate detection and response workflows.

AutomAtED AlERt GEnERAtion & SEvERitY ClASSifiCAtion

- Transforms detected security issues into human-readable alerts through emails with clear severity levels.

CoDE SNiPPEtS

ANN - GEnERiC nEtwoRK AttACKS

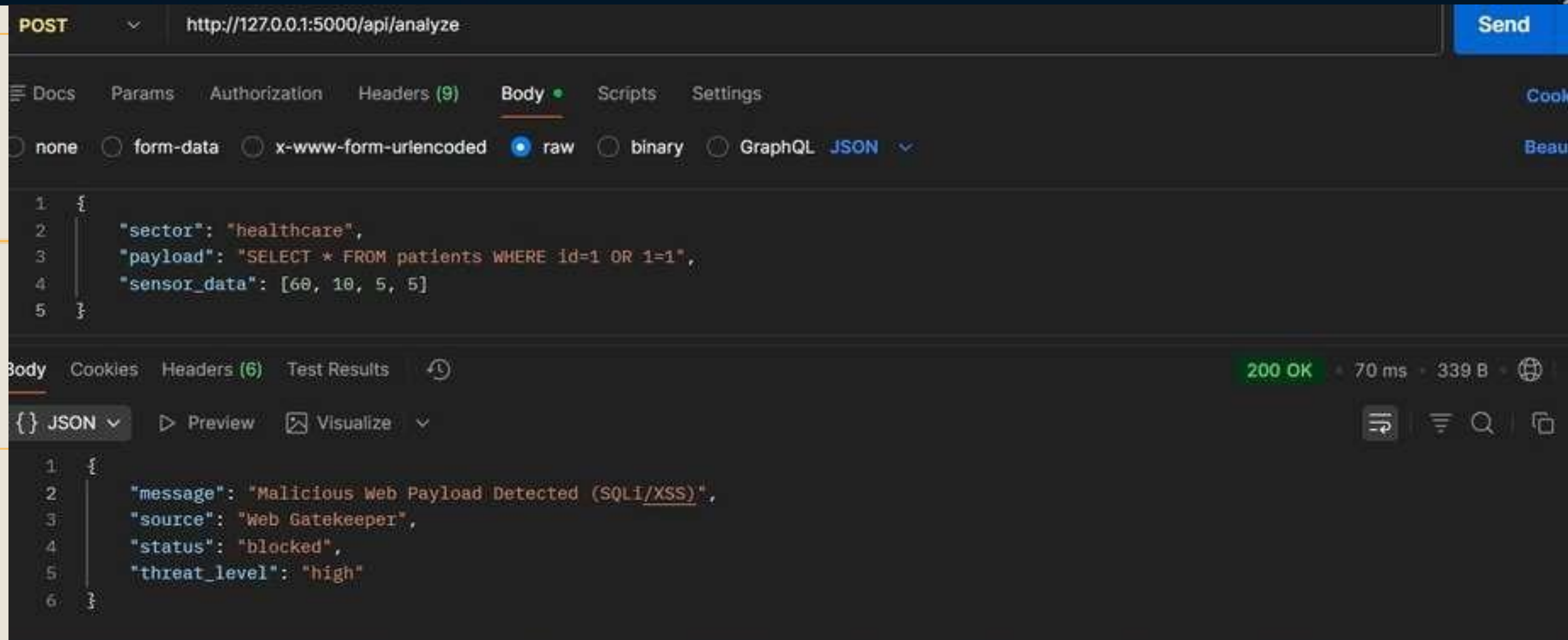
```python
train_idx, _ = train_test_split(range(len(tensor_x)), test_size=0.2)
    loader = DataLoader(TensorDataset(tensor_x[train_idx], tensor_y[train_idx]), batch_size=2048, shuffle=True)

    model = GeneralNetworkShield(input_dim=X.shape[1]).to(DEVICE)
    optimizer = optim.Adam(model.parameters(), lr=0.001)
    criterion = nn.BCELoss()

    model.train()
    for epoch in range(3):
        total_loss = 0
        for X_batch, y_batch in loader:
            X_batch, y_batch = X_batch.to(DEVICE), y_batch.to(DEVICE)
            optimizer.zero_grad()
            out = model(X_batch)
            loss = criterion(out, y_batch)
            loss.backward()
            optimizer.step()
            total_loss += loss.item()
        print(f"   Epoch {epoch+1} Loss: {total_loss/len(loader):.4f}")
```

```python
client_id = data.get('client_id', 'unknown')
    client_weights = data.get('weights')

    if not client_weights:
        print(f"[ERR] No weights in payload from {client_id}")
        return jsonify({"error": "Missing weights"}), 400

    print(f"[INFO] Processing update from {client_id} (Keys: {len(client_weights)})")

    client_state_dict = {k: torch.tensor(v).to(DEVICE) for k, v in client_weights.items()}

    pending_updates.append(client_state_dict)
    print(f"[FL] Received FL update from {client_id}. Progress: {len(pending_updates)}/{REQUIRED_UPDATES_FOR_AVG}")

    if len(pending_updates) >= REQUIRED_UPDATES_FOR_AVG:
        print("[FL] Aggregating Global Model (FedAvg)...")

        # FEDERATED AVERAGING ALGORITHM
        avg_weights = copy.deepcopy(GLOBAL_MODEL.state_dict())
        for key in avg_weights.keys():
            # Stack all updates for this key and take mean
            # Ensure all updates have this key
            valid_updates = [u[key] for u in pending_updates if key in u]
            if valid_updates:
                stacked = torch.stack(valid_updates)
                avg_weights[key] = torch.mean(stacked, dim=0)

        # Update Global Model
        GLOBAL_MODEL.load_state_dict(avg_weights)
        FL_ROUND += 1
        pending_updates = [] # Reset
        print(f"[SUCCESS] Global Model Updated! New Round: {FL_ROUND}")
```

```python
@app.route('/api/fl/submit-update', methods=['POST'])
def submit_update():
```

FEDERAtED LEARninG EnDPoint

PoStmAn moDEl tEStinG



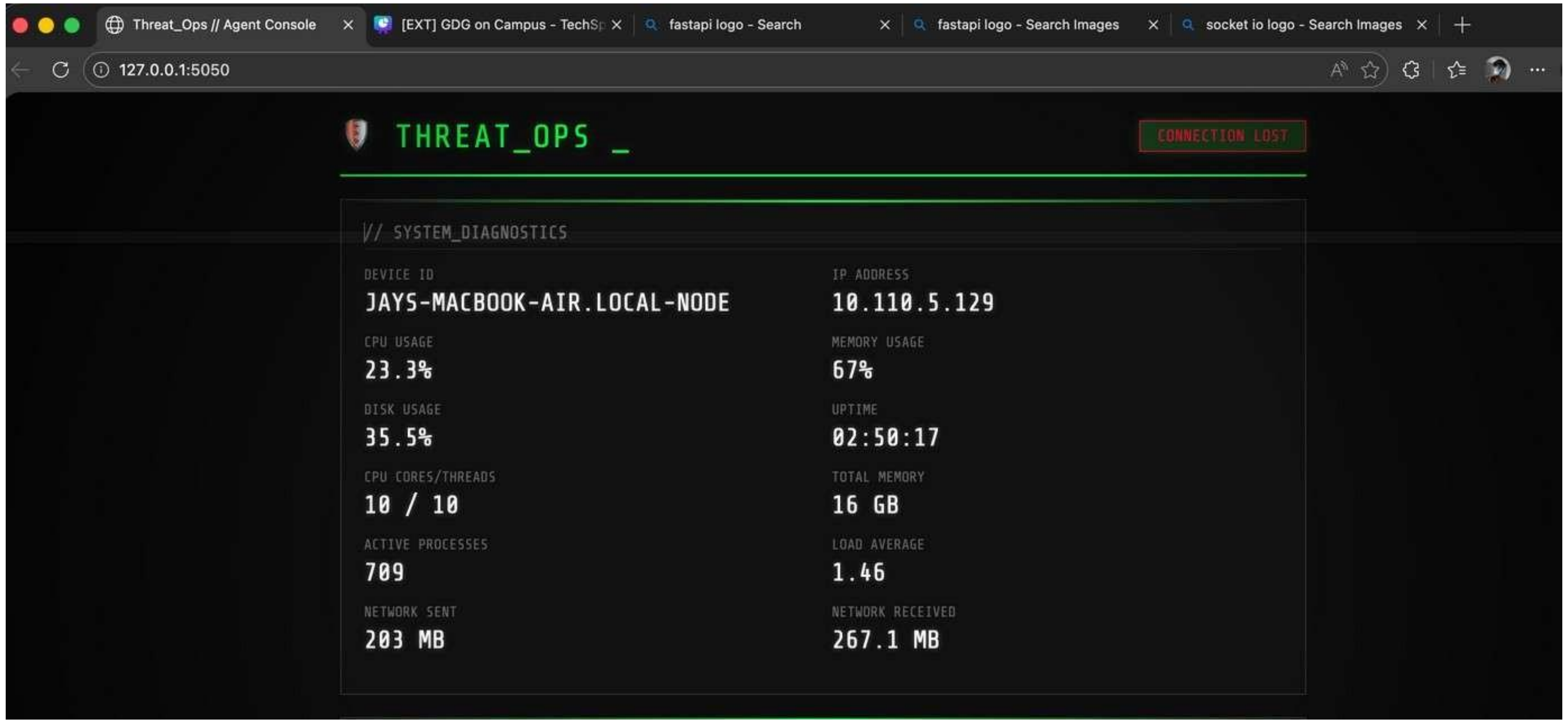
HEAltH CARE ML MODEl

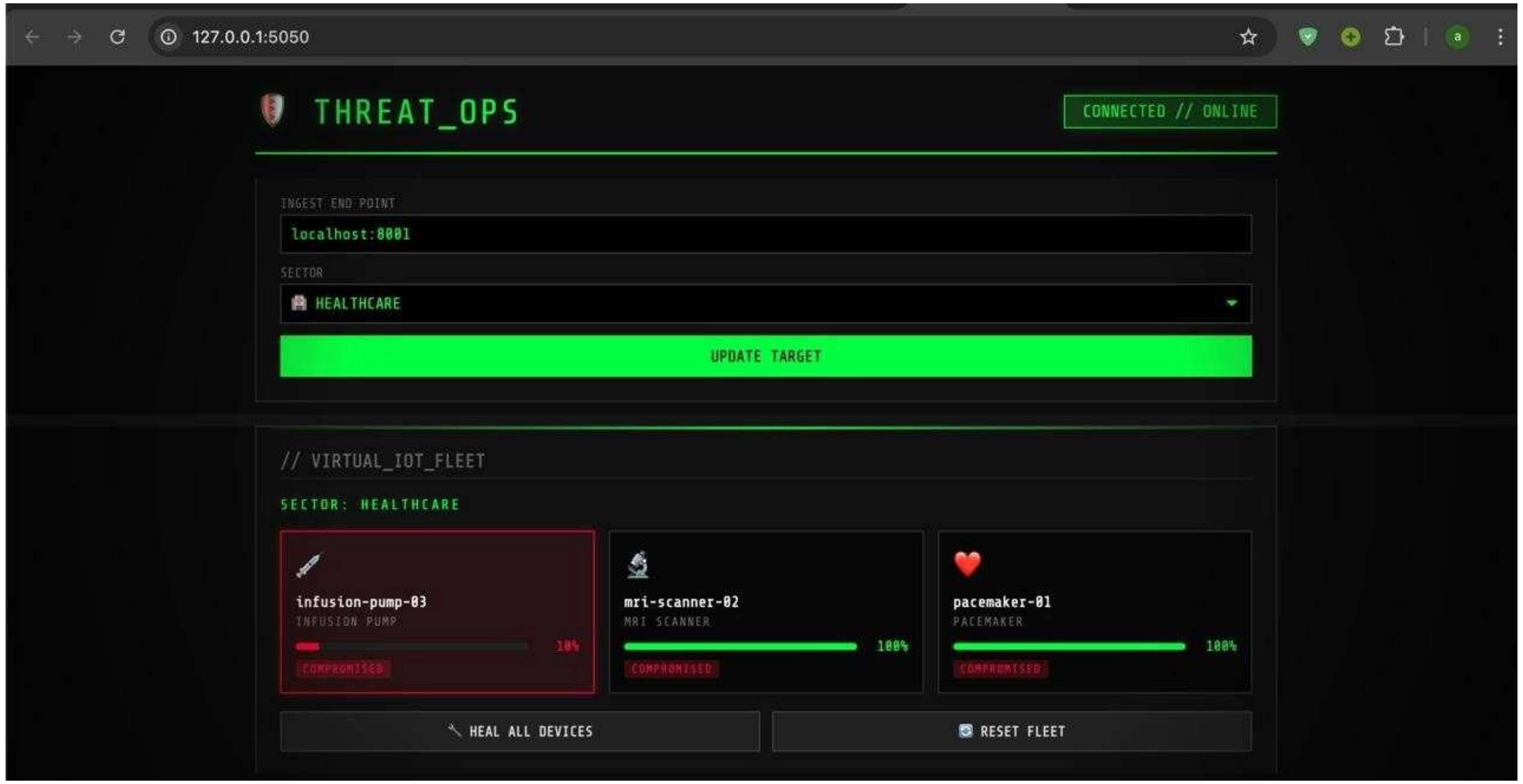
	precision	recall	f1-score	support
Normal	1.00	0.98	0.99	47974
DDoS	0.99	1.00	0.99	51893
accuracy			0.99	99867
macro avg	0.99	0.99	0.99	99867
weighted avg	0.99	0.99	0.99	99867

Accuracy: 0.9910280673295483

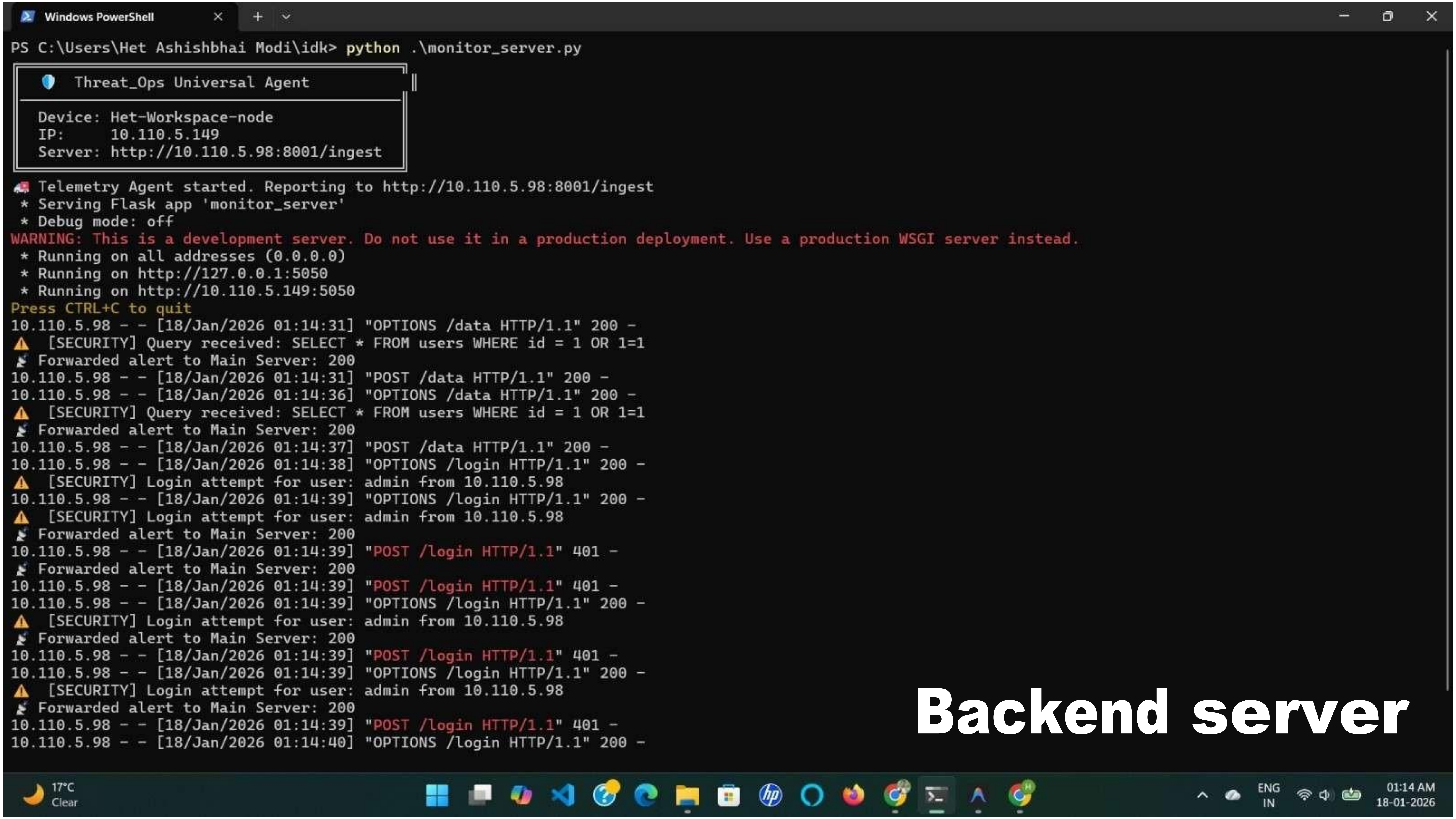
Snapshot of Node software

- **Agent** installed on protected machines - runs as a **background daemon**.
- Auto-discovers device identity and sends **real-time telemetry** to our dashboard.
- **Captures attacks** while monitoring **system health.**
- **Cross-platform support** - same agent works on **Windows, Mac, and Linux**
- Supports **headless** systems, using command line arguments.

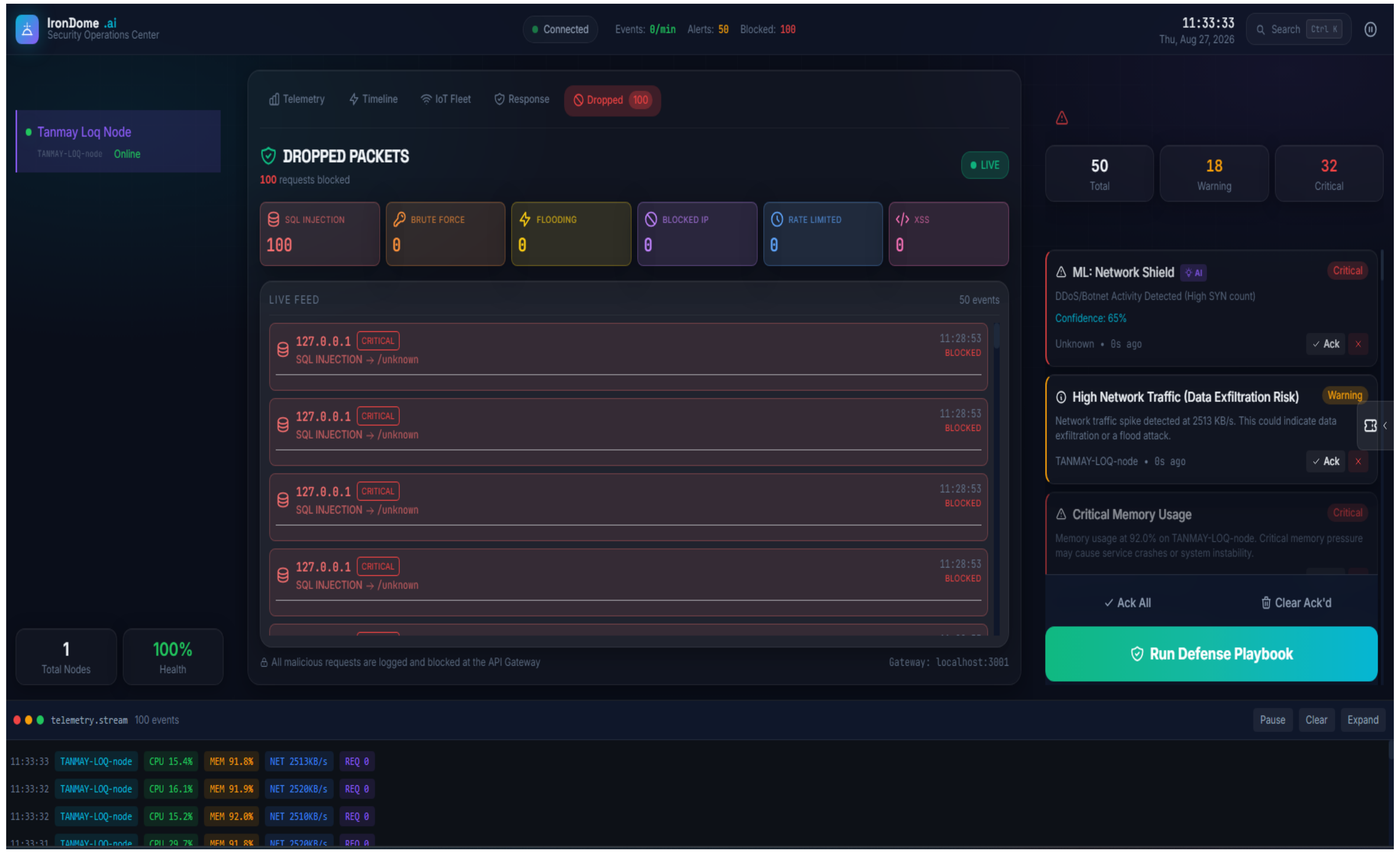


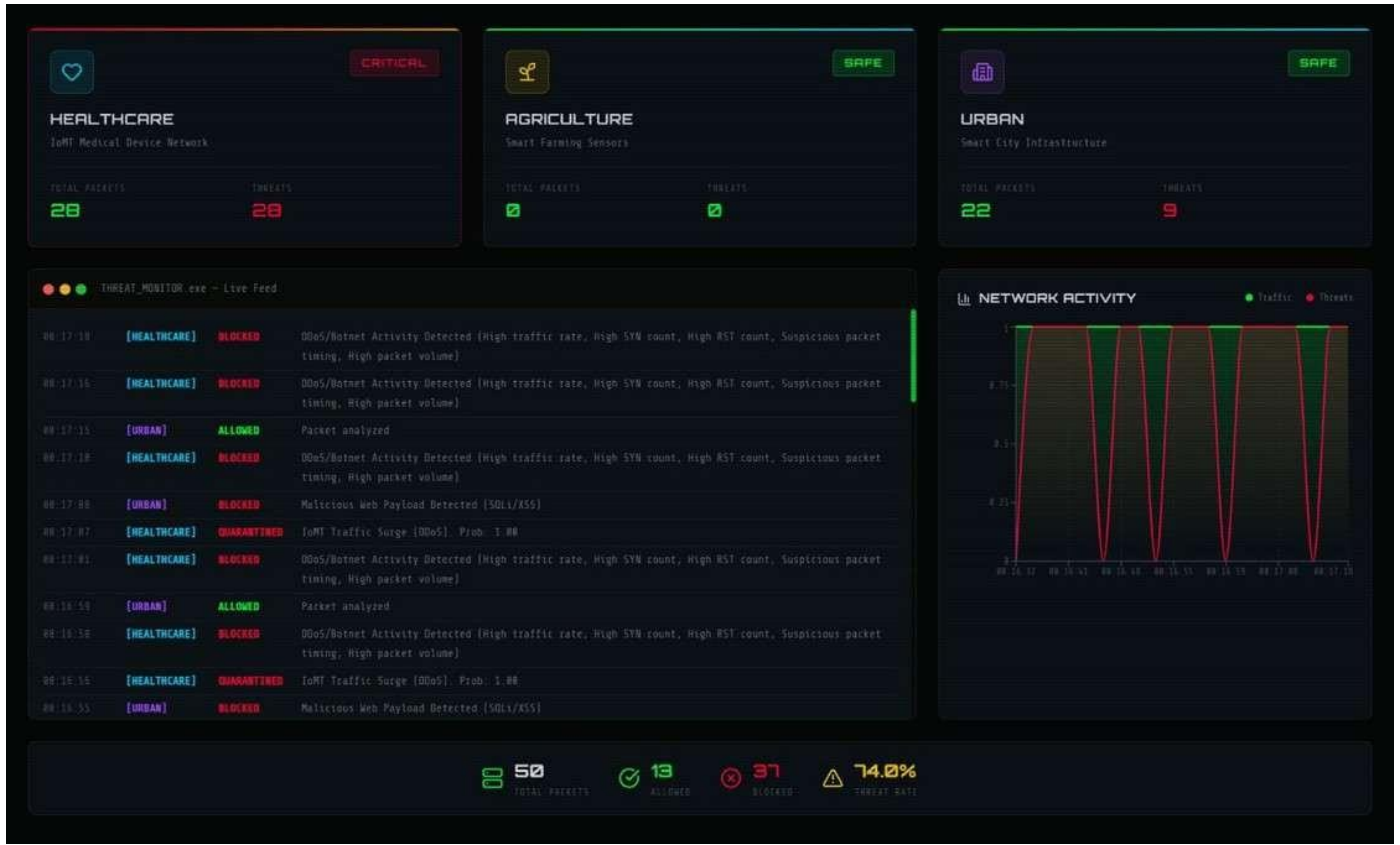
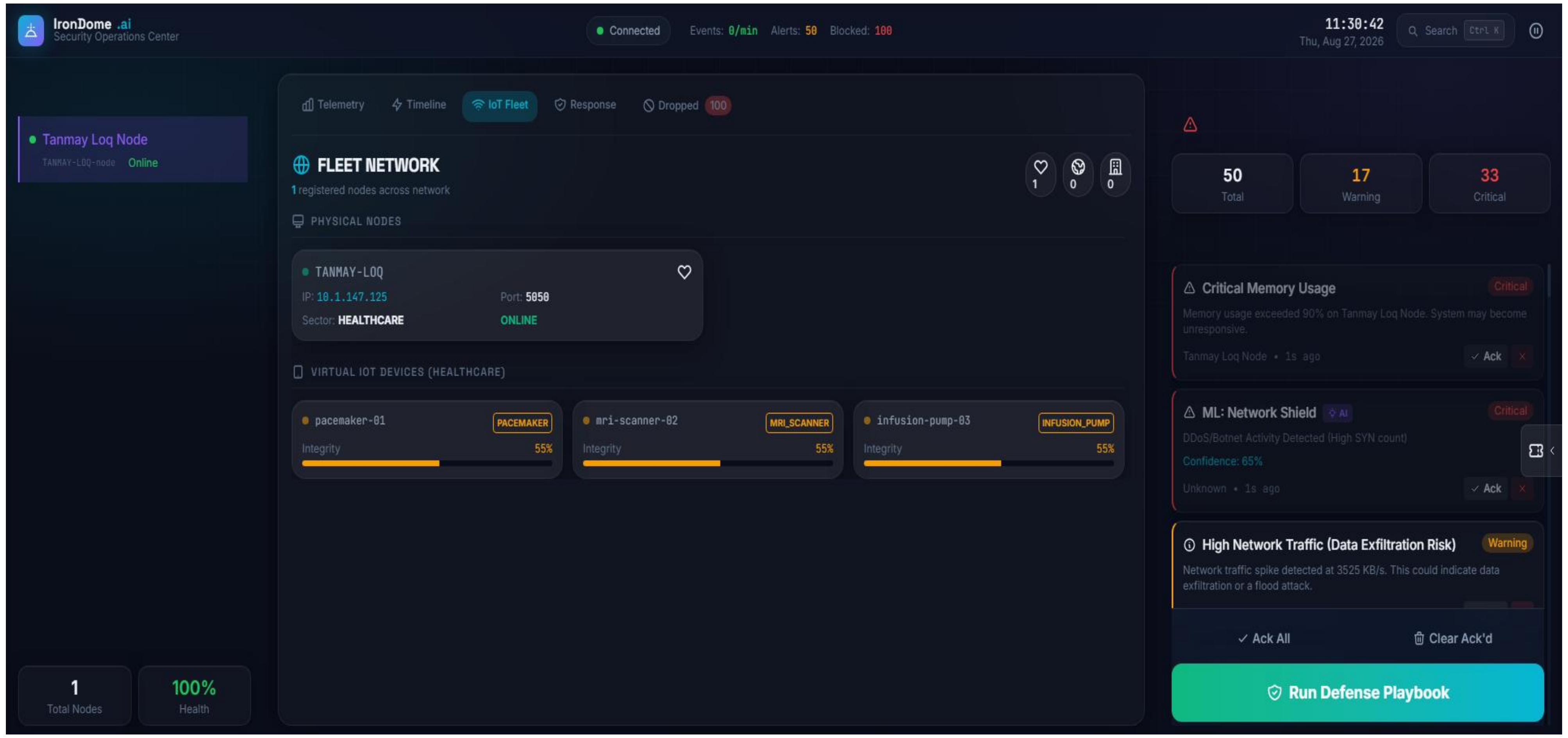


Snapshot of Monitoring dashboard



Backend server


IronDome .ai
Security Operations Center
Connected
Events: 0/min Alerts: 50 Blocked: 100
11:33:33
Thu, Aug 27, 2026
Tanmay Loq Node
Online
Telemetry
Timeline
IoT Fleet
Response
Dropped 100
DROPPED PACKETS
100 requests blocked
LIVE
SQL INJECTION 100
BRUTE FORCE 0
FLOODING 0
BLOCKED IP 0
RATE LIMITED 0
XSS 0
LIVE FEED
50 events
127.0.0.1 CRITICAL
SQL INJECTION → /unknown
BLOCKED
50 Total
18 Warning
32 Critical
ML: Network Shield
Critical
DDoS/Botnet Activity Detected (High SYN count)
Confidence: 65%
Ack
High Network Traffic (Data Exfiltration Risk)
Warning
Network traffic spike detected at 2513 KB/s. This could indicate data exfiltration or a flood attack.
Critical Memory Usage
Ack All
Clear Ack'd
Run Defense Playbook
1 Total Nodes
100% Health
All malicious requests are logged and blocked at the API Gateway
Gateway: localhost:3001
telemetry.stream 100 events
Pause
Clear
Expand

Defense Active!
Defense Active! Executed 6 actions on 6 alerts.
ACTIONS EXECUTED (6)
IP 10.135.48.149 already blocked
Rate limit applied to 10.135.48.149: 10 req/min
IP 10.135.48.149 already blocked
Rate limit applied to 10.135.48.149: 10 req/min
Close

CRITICAL
HEALTHCARE
IoMT Medical Device Network
TOTAL PACKETS 28
THREATS 28
SAFE
AGRICULTURE
Smart Farming Sensors
TOTAL PACKETS 0
THREATS 0
SAFE
URBAN
Smart City Infrastructure
TOTAL PACKETS 22
THREATS 9
THREAT_MONITOR.exe — Live Feed
NETWORK ACTIVITY
50 TOTAL PACKETS
13 ALLOWED
37 BLOCKED
74.0% THREAT RATE

IronDome .ai
Security Operations Center
Connected
Events: 0/min Alerts: 50 Blocked: 100
11:30:42
Thu, Aug 27, 2026
Tanmay Loq Node
Online
Telemetry
Timeline
IoT Fleet
Response
Dropped 100
FLEET NETWORK
1 registered nodes across network
PHYSICAL NODES
TANMAY-LOQ
IP: 10.1.147.125
Port: 5050
Sector: HEALTHCARE
ONLINE
VIRTUAL IOT DEVICES (HEALTHCARE)
pacemaker-01
PACEMAKER
Integrity 55%
mri-scanner-02
MRI_SCANNER
Integrity 55%
infusion-pump-03
INFUSION_PUMP
Integrity 55%
50 Total
17 Warning
33 Critical
Critical Memory Usage
Memory usage exceeded 90% on Tanmay Loq Node. System may become unresponsive.
ML: Network Shield
DDoS/Botnet Activity Detected (High SYN count)
Confidence: 65%
High Network Traffic (Data Exfiltration Risk)
Warning
Network traffic spike detected at 3525 KB/s. This could indicate data exfiltration or a flood attack.
Ack All
Clear Ack'd
Run Defense Playbook
1 Total Nodes
100% Health

ATTACK SIMULATION

TECHNOLOGIES USED

RESOURCES

DATASET LINKS :

- HTTPS://WWW.KAGGLE.COM/DATASETS/PETERFRIEDRICH1/CICDARKNET2020-INTERNET-TRAFFIC
- HTTPS://DATA.MENDELEY.COM/DATASETS/2BW34GHT8B/2
- HTTPS://WWW.KAGGLE.COM/DATASETS/WISAM1985/IOT-AGRICULTURE-2024
- HTTPS://WWW.KAGGLE.COM/DATASETS/HIMADRI07/CICIOT2023/DATA
- HTTPS://WWW.KAGGLE.COM/DATASETS/EVG3N1J/HTTPPARAMSDATASE